Housing Price Prediction: A Comparative Study of Regularization Techniques

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1 Executive Summary

This project investigates linear regression models, enhanced through regularization, to predict housing prices using the Ames, Iowa Housing dataset. We systematically compared standard Linear Regression, Ridge Regression (L2), and Lasso Regression (L1), each trained on two datasets: one with and one without extensive feature engineering. The goal was to evaluate the impact of regularization and feature engineering on prediction accuracy.

The key findings include:

- Ridge Regression with feature engineering was the best model, achieving $R^2 = 0.937$ and RMSE = 0.108.
- Feature engineering reduced test RMSE by up to 10% and increased R^2 by up to 1.6%.
- The most predictive features included "TotalSF_Log", "OverallQual", "GrLivArea_Log", and "Neighborhood_Price".
- Regularization effectively addressed multicollinearity and improved generalization.

The developed models can assist various stakeholders in the real estate market, including homebuyers, sellers, and investors, by providing data-driven estimates of property values. Future work could explore more complex modeling approaches and incorporate additional external data sources to further improve prediction accuracy.

2 Introduction

This project addresses the problem of predicting housing prices in Ames, Iowa, through supervised learning. Specifically, we aim to determine how well linear models with regularization can estimate sale prices, and how feature engineering influences model performance.

To solve this problem, we implemented:

1. Standard linear regression (Ordinary Least Squares)
2. Ridge regression (L2 regularization)
3. Lasso regression (L1 regularization)

Each model was evaluated on both raw and feature-engineered datasets, for a total of six experiments. This approach allowed us to isolate the effect of feature engineering and regularization.

3 Motivation

This project is important because accurate housing price prediction helps various stakeholders make informed decisions. Potential homebuyers can determine if a house is fairly priced, sellers can set competitive prices, and investors can identify properties with growth potential.

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Understanding the factors that influence housing prices can inform urban planning and development strategies.

I'm particularly excited about this project because it provides an opportunity to apply statistical learning techniques to a real-world problem with tangible implications. Real estate is one of the largest markets in the economy, and data-driven approaches can help make this market more efficient and transparent.

Existing questions in this area include:

- Which features have the strongest correlation with housing prices?
- How do different neighborhoods impact property values?
- Can regularization techniques improve prediction accuracy and model interpretability?
- How effectively can we predict housing prices using only physical attributes versus including location data?
- Does feature engineering improve accuracy and interpretability?

4 Method

4.1 Dataset

For this project, I used the Ames Iowa Housing dataset, which contains information on residential properties sold in Ames, Iowa between 2006 and 2010. This comprehensive dataset includes 79 explanatory variables describing various aspects of residential homes. The data is in tabular format with both numerical and categorical features. Key feature categories include:

1. Property characteristics: Square footage, number of rooms, bathrooms, etc.
2. Location information: Neighborhood, zoning classification
3. Quality and condition ratings: Overall quality, exterior condition, etc.
4. Time-related features: Year built, year remodeled, year sold
5. Special features: Fireplaces, pools, garages, etc.

The target variable is SalePrice, which represents the property's sale value in dollars.

4.2 Problem Definition and Feature Space

The problem is defined as a regression task where we predict a continuous value (SalePrice) based on property features. The feature space was refined through:

1. Data preprocessing: Handling missing values, outlier detection, and transformations
2. Feature engineering: Creating new features, encoding categorical variables

4.3 Model Selection

We implemented and compared three models, each with two datasets:

1. Standard Linear Regression: Provides a baseline performance using Ordinary Least Squares (OLS)
2. Ridge Regression: Applies L2 regularization to reduce model complexity and mitigate multicollinearity
3. Lasso Regression: Employs L1 regularization which performs feature selection by shrinking less important feature coefficients to zero

4.4 Implementation Steps

1. Exploratory Data Analysis (EDA):
   - Analyze feature distributions
   - Identify correlations between features and target variable
   - Detect outliers and anomalies
2. Data Preprocessing:
   - Handle missing values through imputation or removal
   - Transform skewed numerical features using log or other transformations
   - Encode categorical variables using one-hot encoding or label encoding
3. Feature Engineering:
   - Create new features that might have predictive power
   - Apply feature scaling to normalize the range of features
4. Model Training:
   - Split data into training and testing sets
   - Train three different linear regression models each with both dataset types
   - Tune hyperparameters using cross-validation
5. Model Evaluation:
   - Assess performance on test set
   - Compare models using various metrics
   - Analyze the improvement in using feature engineering for each of the three models
6. Predictions Visualizations:
   - Analyze predictions of the best performing model
   - Visualize relationships between actual and predicted sales prices
   - Visualize residual distribution between actual and predicted sales prices
7. Feature Importance:
   - Compare feature importance across different models
   - Visualize the most influential features using bar charts
8. Conclusion

4.5 Feature Engineering

Feature engineering was a critical component of our housing price prediction methodology, significantly enhancing model performance. The created features include:

4.5.1 Time-Based Features

- HouseAge, YearsSinceRemodel, IsRemodeled

4.5.2 Area and Size Features

- TotalSF: total square footage
- TotalBath: Weighted sum of bathrooms
- TotalPorchSF: Combined square footage of all porch types

4.5.3 Ratio and Proportion Features

- LotRatio: Calculated as LotFrontage / LotArea
- BedroomRatio: Ratio of bedrooms to total rooms

4.5.4 Interaction and Indicators

- OverallQualCond: Multiplication of OverallQual and OverallCond
- HasPool, HasGarage, HasFireplace

4.5.5 Neighborhood Encoding

- Neighborhood_Price: Mean sale price by neighborhood

4.6 Evaluation Metrics

To evaluate model performance, we used:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R-squared score (R^2)
- Cross-validation scores (5-fold)

5 Results

5.1 Model Performance

After training and evaluating all the models, I found that the regularized models performed better than the standard linear regression. Ridge regression with feature engineering showed the best overall performance with an RMSE of 0.108 and R^2 of 0.937, closely followed by Lasso regression (FE) with similar metrics.

5.2 Performance Metrics Analysis

Model	RMSE (Test)	MAE (Test)	R^2 (Test)	CV RMSE
Linear Regression (No FE)	0.1330	0.0905	0.9043	0.1582
Ridge Regression (No FE)	0.1182	0.0798	0.9245	0.1407
Lasso Regression (No FE)	0.1202	0.0807	0.922	0.1435
Linear Regression (FE)	0.1161	0.0830	0.9270	0.1418
Ridge Regression (FE)	0.1076	0.0749	0.9374	0.128
Lasso Regression (FE)	0.108	0.0755	0.937	0.1269

Table 1: Performance metrics across all models

Feature Engineering Benefits:

- Ridge: $\sim$9% RMSE reduction, +1.4% R^2
- Lasso: $\sim$10% RMSE reduction, +1.6% R^2
- Linear: $\sim$13% RMSE reduction, +2.5% R^2

5.3 Visualizations

The visualizations from the analysis reveal several key insights about housing price prediction in Ames, Iowa:

1. The distribution of sale prices presented by histogram showed that Sales Price was positively skewed. This was addressed with log-transformation.
2. The multicollinearity heatmap and the scatter plots confirmed relationships between sale price and overall quality, garage area, and Total Basement sqft.
3. Feature importance plots highlight that quality/condition metrics (e.g. overall quality, kitchen quality, sale condition) and size-related variables (e.g. TotalSF, GrLivArea) contribute most significantly to price determination.

6 Discussion

6.1 Expected vs. Actual Results

We expected that both Ridge and Lasso regression would outperform OLS due to their ability to handle multicollinearity and prevent overfitting. We anticipated that Lasso might perform particularly well given its feature selection capability in a dataset with numerous features.

Our actual results largely aligned with these expectations. Both regularized models outperformed the standard linear regression, confirming the value of regularization for this problem. However, Ridge regression slightly outperformed Lasso regression. This suggests that feature shrinkage (reducing the magnitude of all coefficients) was more beneficial than feature selection (eliminating some features entirely) for this particular dataset.

The feature importance analysis revealed some surprising insights. While we expected features like overall quality and above-ground living area to be significant predictors, we did not anticipate the substantial impact of basement-related features. The finished basement area emerged as one of the top predictors, indicating that below-ground living space is highly valued in this market.

Additionally, I was surprised by the relatively modest impact of lot size compared to home quality metrics. This suggests that in Ames, Iowa, homebuyers prioritize the quality and condition of the structure more than the size of the land it sits on. This finding contrasts with studies from other real estate markets where land value is often a dominant factor, particularly in high-density urban areas.

6.2 Impact of Feature Engineering

Our feature engineering efforts expanded the feature space from the original set to a significantly larger number of engineered features (multiple new features were created). This process:

1. Captured domain knowledge: By incorporating real estate principles into features like age, total area, and quality-condition interactions.
2. Improved model performance: The R^2 score increased from 0.89 with basic features to 0.94 with engineered features.
3. Enhanced interpretability: Created features that align with intuitive understanding of housing valuation factors.
4. Addressed statistical requirements: Transformations helped meet linear regression assumptions by normalizing distributions and reducing multicollinearity.

6.3 Limitations and Future Work

Despite the strong performance of our models, several limitations should be acknowledged:

1. Temporal relevance: The dataset only includes houses sold between 2006-2010, potentially limiting generalizability to current market conditions, especially considering the significant fluctuations in the housing market since that period.
2. Geographic specificity: The models are trained on data from Ames, Iowa only and may not transfer well to other real estate markets with different valuation dynamics.
3. Missing external factors: Economic indicators, interest rates, and market trends are not captured in the dataset but likely influence housing prices significantly.
4. Model simplicity: While linear models offer interpretability, they may miss complex non-linear relationships between features and price.
5. Limited sample size: Working with a relatively small dataset (especially evident in our limited sample) may not fully capture the diversity of housing characteristics.

To improve this analysis in future work, I would recommend:

1. Feature interaction exploration: Investigate how combinations of features might influence price, such as how the value of square footage differs by neighborhood or how renovation year interacts with original construction year.

2. Advanced modeling techniques: Implement ensemble methods or gradient boosting models to capture more complex patterns and potentially improve prediction accuracy.
3. Time series analysis: Incorporate temporal trends in housing prices to account for market fluctuations and seasonality effects.
4. External data integration: Include economic indicators, school district ratings, crime statistics, and neighborhood development data to provide additional context.
5. Neural network approach: Explore deep learning techniques for potentially higher accuracy, especially for capturing non-linear relationships in the data.
6. Spatial analysis: Incorporate geospatial techniques to better capture location effects beyond simple neighborhood categorization.

7 Conclusion

In conclusion, this project demonstrated the effectiveness of regularized linear regression techniques and feature engineering for housing price prediction. Both Ridge and Lasso provided improvements over standard linear regression, with Ridge regression showing the best overall performance. The analysis highlighted the importance of overall quality, living area size, basement features, and neighborhood in determining housing prices, providing valuable insights for real estate stakeholders, homebuyers, and sellers in the Ames market.

The regularization approach successfully addressed multicollinearity issues common in housing data and prevented overfitting. The interpretability of the linear models allowed for clear identification of key value drivers, making the findings actionable for various stakeholders in the real estate market.

Future research directions could explore more complex modeling techniques and incorporate additional data sources to further enhance prediction accuracy, particularly for luxury properties where the current models showed slightly lower performance.